

4 Recurrences

$$T(n) = 2T(n/2) + n^3 \quad (1)$$

$$\log_b a = \log_2 2 = 1 \quad (2)$$

$$\log^{b^a} = n \quad (3)$$

$$n^3 \in \Omega(n^{\log_b a + 2}) \quad (4)$$

$$\Omega(n^3) \quad (5)$$

$$af(n/b) \leq cf(n) \quad (6)$$

$$2f(n/2) \leq cf(n) \quad (7)$$

$$2 \times \left(\frac{n}{2}\right)^3 \leq cn^3 \quad (8)$$

$$\frac{n^3}{4} \leq cn^3 \quad (9)$$

$$\text{choose } c < 1 \quad (10)$$

$$\text{choose } c = \frac{1}{4} \quad (11)$$

$$T(n) \in \Theta(n^3) \quad (12)$$

5 Correctness

Loop has three parts, init, maintenance, termination.

6 Quicksort